SS1 SECOND TERM MATHEMATICS 2026

OBJECTIVE

1. A sales girl gave a change of #95 .00 to the buyer instead of #100.00. calculate the percentage error.
    a. # 4.00
    b. # 5.00
    c. #2.00
    d. #10.00
    ANSWER: B
2. The length of a book is 1.45m. A boy measured it as 1.90m, find the percentage error to one dp
    a. 31.03
    b. 23.68
    c. 23.7
    d. 31.0
    ANSWER: A
3. Round off 0.045 to 2 dp
    a. 0.04
    b. 0.01
    c. 0.05
    d. 0.02
    ANSWER: C
4. Round off 96495 to the nearest 100
    a. 96410
    b. 96500
    c. 96490
    d. 90000
    ANSWER: B
5. Round off 18145.24 to 3sf
    a. 18200.00
    b. 18100.00
    c. 18100

d. 18200

ANSWER: C

6. Round off 0.00456 to 2 sf
   a. 0.0045
   b. 45
   c. 0.0046
   d. 46

   ANSWER: A
7. Solve $20p^2 - 320 = 0$
   a. P = -4 0r 4
   b. P = -1 or 1
   c. P = 2 or 3
   d. P = 3 or 4

   ANSWER: A
8. Solve the quadratic equation $x^2 + 11x + 18 = 0$
   a. -1 or 2
   b. -2 or -9
   c. 9 or 2
   d. 1 or 2

   ANSWER: B
9. Factorize $2h^2 + 5h + 3$
   a. (2h + 3)(h + 1)
   b. (2h - 3)(h -1)
   c. (4h + 3)(2h - 2)
   d. (h - 4)(3h-2)

   ANSWER: A

Use the below table to answer question 10 – 14

| P | Q | ~Q | P ^~$Q$ | ~ (P^Q) |
|---|---|---|---|---|
| T | T | A | B | T |
| T | F | T | T | F |
| F | T | F | C | D |
| F | F | T | F | E |

10. What is the value of A
    a. T
    b. F
    c. TF
    d. FT

    ANSWER: B

11. What is the value of B
    a. T
    b. F
    c. TF
    d. FT

    ANSWER: B

12. What is the value of C
    a. T
    b. TF
    c. FT
    d. F

    ANSWER: D

13. What is the value of D
    a. TF
    b. FT
    c. F
    d. T

    ANSWER: D

14. What is the value of E
    a. T
    b. F
    c. TF
    d. FT

    ANSWER: A

15. Find area of a circle with radius 21cm
    a. 1386cm2
    b. 345cm2

c. 247cm2
d. 1456cm2

ANSWER: A

16. Find the area of a square with one side 4.5cm
    a. 16.25cm$^2$
    b. 18.75cm$^2$
    c. 20.25cm$^2$
    d. 21.50cm$^2$

    ANSWER: C

17. Find the area of a square field with each diagonal 150m long
    a. 32,500m
    b. 11,250m$^2$
    c. 22,500m$^2$
    d. 22,500

    ANSWER: B

18. The area of a circle is 81$\pi$cm2. Find the diameter of the circle.
    a. 7cm
    b. 9cm
    c. 13cm
    d. 18cm

    ANSWER: D

19. Calculate the circumference of a circle, centre o and radius 3.5cm($\pi = 22/7$).
    a. 11cm
    b. 15cm
    c. 13cm
    d. 22cm

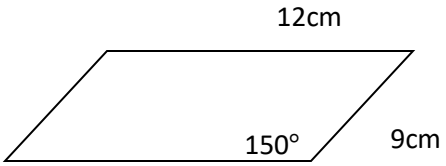
    ANSWER: D

20. The perimeter of a parallelogram is 38cm. if one of the sides is 12cm, find the other side
    a. 7cm
    b. 9cm
    c. 13cm

d. 6cm

ANSWER: A

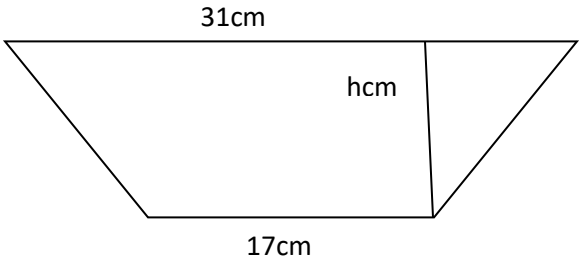
21. Calculate the area of the figure below




a. 1.5cm2
b. 10.5cm2
c. 27cm2
d. 54cm2

ANSWER: D

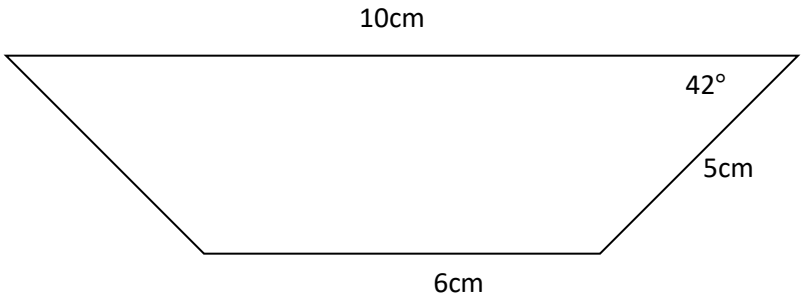
22. The trapezium below has an area of 456cm$^2$, calculate the distance between its parallel sides.




a. 29cm
b. 27cm
c. 19cm
d. 17cm

ANSWER: C

Use the diagram below for 23 – 24



23. Calculate the perpendicular distance between the parallel sides
    a. 3.35cm
    b. 3.72cm
    c. 4.50cm
    d. 4.62cm
    ANSWER: A
24. Calculate correct to the nearest $cm^2$ the area of the trapezium
    a. $27cm^2$
    b. $30cm^2$
    c. $36cm^2$
    d. $37cm^2$
    ANSWER: A
25. Calculate the area of a circle with diameter 8m to the nearest m2
    a. 50m2
    b. 40m2
    c. 30m2
    d. 20m2
    ANSWER: A
26. Factorise $V^2 – 6v – 27$
    a. (v-9)(v+3)
    b. (v-9)(v-3)
    c. (v + 3)(v-2)
    d. (v-2)(v-3)
    ANSWER: A
27. Solve the $3t^2 – 12t = 0$
    a. t = 1 0r 2
    b. t = 0 or 2
    c. t = 0 or 4
    d. t = -4 or 0
    ANSWER: C
28. Solve the equation $3x^2 – 75 = 0$

a. x = -5 or 5
b. x = 5 0r 5
c. x = 1 or 2
d. x = 2 or 2

ANSWER: A

29. Round of 6.0964 to the nearest whole number
    a. 1
    b. 6
    c. 7
    d. 5

    ANSWER: B

30. Estimate 8.4 x 9.6 to 1sf
    a. 80
    b. 60
    c. 70
    d. 50

    ANSWER: A

31. Estimate 99.9 x 313 to one significant
    a. 30
    b. 300
    c. 400
    d. 40

    ANSWER: B

32. Estimate The value of $\frac{75.2 \times 120}{4.162 \times 49.6}$ correct to one significant figure
    a. 20
    b. 30
    c. 40
    d. 60

    ANSWER: C

Use the table below to answer question 33 – 38

| P | Q | (PvQ) | (P^Q) | (PvQ)V(P^Q) |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| T | T | T | T | T |
| T | F | A | B | T |
| F | T | T | C | D |
| F | F | F | F | E |

33. What is the value of A
    a. T
    b. F
    c. TT
    d. FT

    ANSWER: A

34. What is the value of B
    a. F
    b. T
    c. FT
    d. TF

    ANSWER: A

35. What is the value of C
    a. T
    b. F
    c. TF
    d. FF

    ANSWER: B

36. What is the value of D
    a. T
    b. F
    c. TT
    d. FF

    ANSWER: B

37. The following are special angles except----------
    a. 30°
    b. 60°
    c. 90°
    d. 25°

D

38. An arc subtends an angles of 105 at the centre of a circle of radius 6cm. find the length of the arc if $\pi$ = 22/7
    a. 11cm
    b. 12cm
    c. 13cm
    d. 14cm

    ANSWER: A

39. Calculate the perimeter of a sector of a circle of radius 7cm, the angle of the sector being 108° if $\pi$ is 22/7
    a. 13.2cm
    b. 40cm
    c. 5cm
    d. 27.2cm

    ANSWER: D

40. An arc subtends an angle of 72 at the circumference of a circle of radius 5cm. calculate the length of the arc in terms of $\pi$
    a. $\pi$
    b. 2 $\pi$
    c. 3 $\pi$
    d. 4 $\pi$

    ANSWER: D

MATHEMATICS

SECTION B

ANSWER NO 1 AND ANY OTHER 4

1) Copy and Complete the table of values for the relation $y = x^2 - 2x - 1$ for $-2 \leq x \leq 4$

| x | -2 | -1 | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|---|---|
| y | | | | | | | -4 |

   a. Using scales of 2cm to 1 units on both axis
   b. From the graph Find the minimum value of y
   c. Use the graph to find the roots of the equation $x^2 - 2x - 1 = 0$
   d. Using the axis draw graph of $y = 2x - 3$
   e. From the graph determine the roots of the equation $x^2 - 2x - 1 = 2x - 3$

2) Solve $x^2 = 10x + 39$

   b) factorise b2 – 12b – 13

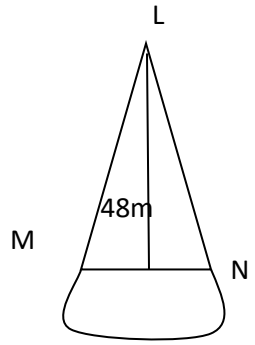
3) Estimate $\frac{95.4 \times 149}{5.2 \times 49.1}$ correct to one significant figure.

   B) A pencil is 18cm long someone estimated it length to be 20cm. calculate the percentage error

4) Show that ~(P^Q) v Q is tautology

   b) Show that P^(Q^P)

5)




a. The diagram shows a wooden structure in form of a cone, mounted on a hemispherical base. The vertical height of the cone is 48m and the base radius is 14m.[take $\pi$ 22/7]
b. What is the length of an arc which subtends an angle of 60 at the center of a circle of radius 7

6) Copy and complete the table of values for the relation $-x^2 + x + 2$ for -
-$3 \leq x \leq 3$

| x | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|---|
| y | -4 | | | | | | -4 |

a. Using scales of 2cm to I unit on x axis and 2cm to 2units on y axis
b. From the graph find:
   I. Maximum value of y
   II. The roots of the equation $x^2 - x - 2 = 0$
   III. Gradient of curve at x = - 0.5

7) a) What is the total surface area of a closed cylinder of height 10cm and diameter 7cm? (take 22/7) (6marks)
b) Find the volume of the cylinder(4marks)